

ShiftSleep-UQ Supplementary Material

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This supplement is a structured reproducibility companion to the first submission candidate. Every numerical table and diagnostic below is reproduced from frozen Step 17 artifacts; no new training, inference, calibration fit, bootstrap, or statistical test is introduced here. Repository-relative paths are included so the rendered supplement remains auditable.

S1. Dataset and channel details

The benchmark uses Sleep-EDF Expanded Sleep Cassette recordings and ISRUC Cohort I. The frozen accounting is 78 Sleep-EDF subjects (153 recordings; 414,961 valid epochs) and 100 ISRUC subjects, of which 99 are included after excluding one unsupported layout (89,312 valid epochs). The input channels are EEG Fpz-Cz and horizontal EOG for Sleep-EDF. ISRUC montage families are ISRUC_A1A2 and ISRUC_M1M2. Labels are harmonized to Wake, N1, N2, N3, and REM; R&K stages 3 and 4 map to N3. Movement, unknown, artifact, and unscored epochs are excluded.

Table 1: Frozen benchmark design; counts are authoritative subject-role allocations.

Direction	Source	Target	TRAIN	DEV	CAL	TEST	Target cohort
D1	Sleep-EDF	ISRUC	47	12	8	11	99
D2	ISRUC	Sleep-EDF	59	15	10	15	78

S2. Architecture and training

B0 is a dual raw-signal branch convolutional model with independent EEG and EOG branches, GroupNorm, GELU, residual blocks, adaptive pooling, 128-dimensional branch embeddings, modality masking, concatenation, and classification (654,597 parameters). B1 uses the same architecture and training protocol, exposing source TRAIN to full input with probability 0.50, EEG-only input with probability 0.25, and EOG-only input with probability 0.25. The frozen methods map is `reports/step17_methods_evidence_map.md`; checkpoint and configuration manifests remain in the repository evidence package.

S3. Full per-seed results

The complete frozen per-seed B0 and B1 tables are retained as CSV artifacts and are not recomputed here.

`reports/paper_supplement/all_b0_v1.2_results.csv`

`reports/paper_supplement/all_b1_results.csv`

`reports/paper_supplement/b1_vs_b0_paired_results.csv`

The compact canonical B0 primary table is reproduced below.

Dir.	Cond.	macro-F1	NLL	AURC	$ \Delta_{cov} $
D1	C0	.74800	.22027	.00875	.09958
D1	C1	.66336	.28017	.01338	.09854
D1	C2	.46627	.67956	.09406	.09342
D1	C3	.59490	1.09423	.18813	.07985
D1	C4	.50639	1.11727	.25705	.07859
D1	C5	.28955	1.72668	.60266	.03810
D2	C0	.73072	.63558	.08865	.06561
D2	C1	.64629	.79619	.13442	.08528
D2	C2	.48478	1.18798	.32971	.06986
D2	C3	.48354	1.10896	.16343	.00884
D2	C4	.38231	1.60744	.32362	.02424
D2	C5	.36043	1.26055	.45743	.01953

S4. Stage analysis

Stage-level B1–B0 recall changes for Wake, N1, N2, N3, and REM across C3–C5 are retained in `reports/paper_supplement/stage_level_effects.csv`; the complete frozen source is unchanged.

S5. Confusion matrices

Frozen confusion changes are retained in `reports/paper_supplement/confusion_changes.csv`. No confusion matrix was regenerated for Step 19.

S6. Interaction analysis

The frozen interaction contrasts are retained in `reports/paper_supplement/interaction_contrasts.csv`. These are deterministic copies of the Step 17 evidence package, not new contrasts.

S7. ISRUC montage sensitivity

The frozen montage sensitivity analysis is retained in `reports/paper_supplement/montage_sensitivity.csv`. The main text treats A1A2 versus M1M2 as a dataset-specific limitation and does not infer montage causality.

S8. Source-mask calibration

The source-only mask-specific temperature and adaptive prediction-set diagnostics are retained in `reports/paper_supplement/source_mask_calibration.csv`. They are secondary controls and do not authorize a learned reliability method.

S9. Oracle diagnostics

Target-label oracle diagnostics are retained in `reports/paper_supplement/oracle_diagnostics.csv`. They are post-hoc diagnostic upper bounds and are not deployment evidence.

S10. B0 v1.2 correction

The historical correction audit is retained in `reports/paper_supplement/b0_v1.1_to_v1.2_repair_audit.csv`. B0 v1.2 is authoritative; the earlier artifact remains preserved for auditability.

S11. Reproducibility and hashes

The authoritative evidence manifest is `reports/paper_evidence/paper_evidence_manifest_v1.json`. The manuscript-specific figure source audit is `reports/step19_figure_source_audit.csv`; the Figure 2 binding audit is `reports/step19_fig2_source_audit.csv`. The repository contains the deterministic synthesis script `scripts/step17_paper_synthesis.py`. Raw datasets, prediction bundles, checkpoints, heavy bootstrap shards, and credentials are not redistributed.

Frozen non-claims

The study does not establish universal uncertainty failure, universal modality-dropout superiority, montage causality, clinical safety, arbitrary-shift conformal validity, target-free recovery from all shifts, a novel calibration method, state-of-the-art sleep staging, SHHS validation, a new architecture, or performance on unevaluated populations.